
Snap-HyRE: When Generated Legal Queries Help Legal RAG

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Abstract

Legal RAG often fails before the answer model sees useful law: the question is a poor search query. We study Snap-HyRE, a fixed two-call method that first writes a private draft answer and a separate legal search passage, retrieves with only that passage, and then answers from retrieved evidence. On BarExamQA, Snap-HyRE improves over raw-question RAG for all three model sizes, with gains up to +5.1 percentage points and consistent Hit@5 gains. HousingQA shows the boundary condition: jurisdiction filtering raises raw Hit@5 from 2.8% to 36.9%, after which raw state-filtered retrieval is a strong baseline. The results support a narrow claim: draft-guided legal search passages help in weak-query regimes, and legal RAG papers should report answer accuracy together with whether the right legal evidence was retrieved.

1. Introduction

Legal RAG should work well for legal question answering: many answers turn on statutes, regulations, holdings, or compact doctrinal rules, and retrieval can put that text in front of a language model (Lewisa et al., 2020). The hard part is often the first query. A bar-exam fact pattern may mention a driver, crutches, and a banana peel, while the support passage states a black-letter rule about proximate cause. A housing question may name a state, while an unscoped national corpus can return a plausible statute from the wrong jurisdiction. Final answer accuracy alone hides these failures.

Generated-query retrieval attacks this query mismatch. HyDE embeds a hypothetical document rather than the user’s original query (Gao et al., 2023); query ex-

pansion and rewriting use related generated search text (Wang et al., 2023). These methods are attractive in law because a model can often infer the legal category behind a fact pattern before retrieval sees the corpus. They can also fail by abstracting away anchors, choosing the wrong legal frame, or retrieving plausible but irrelevant authority.

We introduce Snap-HyRE, a fixed two-call generated-query method. The first model call writes two objects: a draft answer and a separate legal search passage. The draft answer is private. It can guide the vocabulary of the search passage, but it is not embedded, retrieved, or shown to the final answer call. The second call answers the original question from the retrieved legal text. The method uses model priors for query formation while keeping answer selection tied to corpus evidence.

We report a two-benchmark study. BarExamQA tests the weak-query regime, where fact-heavy questions need to be translated into rule-shaped retrieval queries. HousingQA tests the corpus-interface regime, where retrieval is not meaningful until every method searches only the jurisdiction named in the question.

The main result is sharpest on BarExamQA. Across Llama 3.1 8B, Gemma 4 26B, and Llama 3.3 70B, Snap-HyRE improves raw-question RAG by +2.4, +4.0, and +5.1 percentage points. It also raises BarExamQA Hit@5 from 1.4% for raw-question retrieval to 9.5–12.1% for the three Snap-HyRE generators. HousingQA gives a different lesson: adding the jurisdiction state filter raises raw Hit@5 from 2.8% to 36.9%, making state-filtered raw retrieval a strong baseline rather than a method to beat by default.

This paper makes three contributions: it defines Snap-HyRE as a fixed two-call legal RAG method; it compares raw-question RAG, HyDE, Snap-HyRE, and oracle evidence controls on BarExamQA and HousingQA; and it analyzes when the method helps by reporting both answer accuracy and gold evidence in the top five retrieved passages. Structured query rewriting and additional diagnostics are reported in the appendix.

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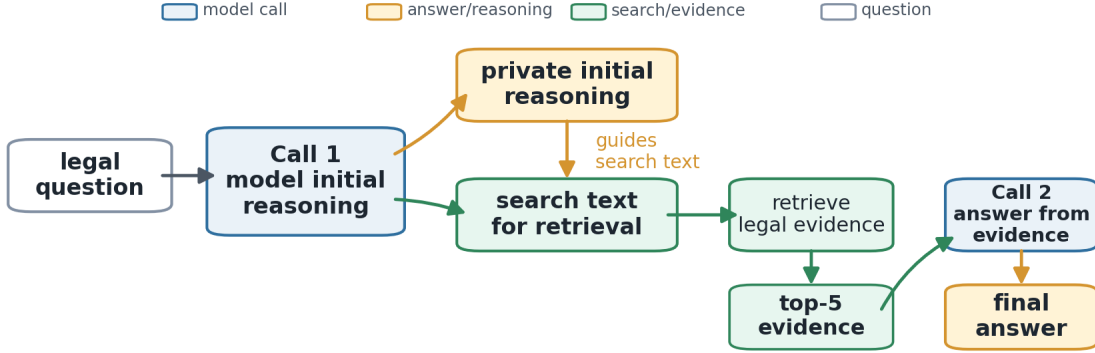


Figure 1. Snap-HyRE (ours). Call 1 drafts privately and writes search text; only the search text is retrieved. Call 2 answers from the original question and retrieved evidence.

2. Why Legal Queries Fail

Generated search text. RAG combines a generator with an evidence store (Lewis et al., 2020). Its legal promise is simple: retrieve the relevant rule or statute, then answer from it. But the query and the corpus often use different language. HyDE addresses this by generating a hypothetical document and embedding that text for retrieval (Gao et al., 2023). Query2doc and related expansion methods similarly use a language model to turn sparse query text into richer retrieval handles (Wang et al., 2023). For legal tasks, the useful generated text is usually not a longer version of the question. It is a passage in the style of the authority the retriever should find.

Legal RAG evaluation. Legal NLP benchmarks and corpora have standardized many legal-understanding tasks (Guha et al., 2023; Chalkidis et al., 2022; Henderson et al., 2022), and recent work has emphasized the risk of unsupported legal assistant answers (Magesh et al., 2024). Legal RAG benchmarks such as LegalBench-RAG and Legal RAG Bench argue for evaluating retrieval and answering together (Pipitone & Alami, 2024; Butler & Butler, 2026). We follow that discipline but keep the methods fixed: every row uses a predetermined retrieval interface, and the analysis asks whether failure comes from missing evidence or from using retrieved evidence poorly.

Two failure modes. BarExamQA and HousingQA expose different legal retrieval problems. BarExamQA questions are fact patterns whose support passages often state compact legal rules. HousingQA questions are jurisdiction-specific statutory questions over a national corpus. The first setting tests whether generated legal passages can translate facts into doctrine.

The second tests whether the corpus has been scoped to the right jurisdiction before method comparisons begin.

3. Snap-HyRE

3.1. Design Rationale

Snap-HyRE makes a narrow design bet: a model’s first instinct can be useful for search, but unsafe as evidence. Legal questions often hide the useful retrieval target behind facts. A model may recognize that target before seeing the corpus: proximate cause, a hearsay exception, a notice rule, or a state-specific statutory remedy. Snap-HyRE uses that prior to write a legal search passage, then forces the final answer to come from retrieved text.

This gives the method two guardrails. First, the initial answer is private. It can shape the search passage, but it is not passed to the retriever as an answer label and is not shown to the final answer call. Second, the search passage is written like legal authority rather than like chain-of-thought reasoning: it names the rule or statutory condition that would resolve the question.

3.2. What We Compare

The main comparison keeps the retrieval and answer interface fixed within each dataset. All non-oracle retrieval rows answer from $k = 5$ passages. The gold passage row is an evidence-use control, not a deployable retrieval method.

3.3. Two Calls, One Retrieval Query

Snap-HyRE separates query formation from final answering. Given a formatted legal question q , the first call produces a private draft answer a_0 and a legal

Table 1. Methods in the main comparison. The retrieval-query column names what is embedded or otherwise used to find context; the answer-context column names what the final answer call receives.

Method	Calls	Retrieval query	Answer context	Purpose
LLM only	1	none	original question	Parametric baseline.
Raw question RAG	1	raw question text	question plus top- k evidence	Baseline retrieval interface.
HyDE	2	generated legal passage	question plus evidence retrieved by that passage	Generated prose without a draft-answer prior.
Snap-HyRE (ours)	2	draft-guided legal search passage	question plus evidence retrieved by that passage	Proposed method; the draft answer is private.
Gold passage	1	none; labeled evidence is supplied	question plus gold passage	Oracle control for evidence use.

search passage p :

$$(a_0, p) = G_{\theta}^{\text{snap}}(q). \quad (1)$$

The retriever embeds only p and returns top- k evidence $E_k = R_k(p)$. The final answer is then

$$\hat{a} = G_{\theta}^{\text{answer}}(q, E_k), \quad (2)$$

with a_0 withheld from the final answer context. The first call can use model priors to choose legal vocabulary, but the second call must answer from the retrieved corpus text.

The retrieval passage is constrained to be short and reference-like. It should describe the controlling rule, statutory condition, legal relationship, or doctrinal test without answer labels or an Answer: header. Outputs that omit the required draft answer or clean passage are excluded from reported rows rather than silently replaced with raw-question retrieval.

The difference from HyDE is narrow. HyDE writes a legal passage directly from the question. Snap-HyRE first forms a tentative legal answer and uses that private judgment only to decide what kind of passage to search with. For a torts fact pattern about a broken leg, crutches, and a later fall, this can turn an accident-heavy query into a passage about proximate cause and foreseeable subsequent injury.

3.4. Keeping Retrieval Interfaces Fair

BarExamQA retrieval uses the shared fact pattern and question stem; generated methods may see option text, but no correctness marker. HousingQA retrieval uses the question text under a jurisdiction state filter derived from the dataset metadata. That filter is not a Snap-HyRE feature. It is the benchmark interface

Table 2. Main benchmark roles.

Benchmark	Retrieval challenge	Role in the paper
BarExamQA	Fact patterns often embed like factual narratives while support passages state compact legal rules.	Primary test of whether generated legal search passages improve raw retrieval.
HousingQA	Questions name a jurisdiction, but the corpus contains statutes from many jurisdictions.	Interface test: all retrieval methods must use the same state filter before rankings are meaningful.

used by every HousingQA retrieval row in the main paper.

A structured rewrite baseline is kept as an appendix control. It is useful for checking whether generic query rewriting explains the gains, but it is not the main contrast: the paper asks whether a private draft answer helps generate a better legal retrieval passage than HyDE or the raw question.

4. Evaluation

4.1. Two Legal Retrieval Regimes

The main paper reports two exact-scored legal QA benchmarks.

BarExamQA. BarExamQA contains 1,195 multiple-choice bar-exam questions in the evaluation split used here. The retrieval corpus is a frozen support-passage collection with official gold passage ids. Hit@5 and MRR@5 therefore measure whether the retriever exposes the labeled rule passage before the answer model chooses an option.

HousingQA. HousingQA contains 6,853 yes/no statutory entailment questions over state housing statutes. The corpus contains many jurisdictions, and each question names the relevant state. All main HousingQA retrieval rows apply that state as a metadata filter before ranking. Rows that do not use this filter are excluded from the main HousingQA comparison.

4.2. Models, Baselines, and Retrieval Stack

We evaluate three hosted model configurations: Llama 3.1 8B Instant (groq-llama8b), Gemma 4 26B through OpenRouter (or-gemma4-26b), and Llama 3.3 70B Versatile (groq-llama70b). The main deployed comparisons are LLM-only, raw question RAG, HyDE, and Snap-HyRE. Gold-passage rows are oracle controls for evidence use. Structured rewrite and gold-plus-neighbor diagnostics appear in the appendix.

All non-oracle retrieval rows answer from $k = 5$ retrieved passages. Runs use seed 42, temperature 0, strict no-silent-fallback execution, and a 2048-token answer cap unless a row note says otherwise. Retrieval uses Chroma collections built with Alibaba-NLP/gte-large-en-v1.5 embeddings and cross-encoder/ms-marco-MiniLM-L-6-v2 reranking before top- k truncation.

Generated-query rows are evaluated in two stages. First, the query-generation call writes the HyDE passage or Snap-HyRE search passage. Then retrieval stores the ranked document ids. The answer run replays those ids strictly; an invalid generated passage or missing cache entry is not replaced with raw-question retrieval. This keeps the comparison about the method’s actual search text.

4.3. Answer and Evidence Metrics

The downstream metric is exact answer accuracy. BarExamQA uses the last valid multiple-choice answer line, and HousingQA uses Answer: Yes or Answer: No. Retrieval is measured by Hit@5 and MRR@5: whether any labeled support passage appears in the top five, and how early it appears.

We also report efficiency as logical LLM calls, answer-stage tokens per question, scored-answer calls per question, and correct answers per million answer-stage tokens. These values do not make two-call methods free: the first-stage query-generation call is counted in the logical method footprint and logged separately. They show how much context the final answer call sees once retrieval has selected evidence.

The main tables include only validated full rows. Missing cells are not imputed. Appendix A.1 lists the in-

clusion rule, coverage grid, supplemental controls, top- k diagnostics, efficiency table, and reproducibility details.

5. Results

5.1. When Snap-HyRE Helps Answers

BarExamQA is the clearest positive setting for Snap-HyRE. Raw-question RAG is a weak retrieval handle: its Hit@5 is 1.4%. Snap-HyRE raises Hit@5 to 9.5%, 12.1%, and 11.1% across the 8B, 26B, and 70B model slices. Those retrieval gains convert to final answer gains over raw-question RAG: 54.5% to 56.9% for Llama 3.1 8B, 78.0% to 82.0% for Gemma 4 26B, and 74.6% to 79.7% for Llama 3.3 70B. Averaged across the three BarExamQA model slices, Snap-HyRE reaches 72.9% accuracy, compared with 69.0% for raw-question RAG and 72.2% for HyDE.

HousingQA tells a different story. The unfiltered national corpus made retrieval mostly a wrong-jurisdiction problem; adding the state filter raises raw Hit@5 from 2.8% to 36.9%. On the included state-filtered rows, raw-question RAG is strong: 62.3% for 8B and 62.1% for 70B. HyDE reaches 59.1% on the 8B row, and Snap-HyRE reaches 59.6% on the 70B row. This is a useful negative boundary: if the question already names the jurisdiction and issue, state-filtered raw retrieval can be hard to improve.

5.2. Did the Right Evidence Appear?

Table 4 separates retrieval from answer selection. On BarExamQA, generated legal passages improve exact evidence exposure: raw retrieval averages 1.4% Hit@5, HyDE averages 10.0%, and Snap-HyRE averages 10.9%. MRR@5 moves in the same direction, from 0.7% for raw-question retrieval to 5.0% for HyDE and 5.4% for Snap-HyRE.

On HousingQA, the state filter is the dominant retrieval correction. Raw state-filtered retrieval reaches 37.0% Hit@5, while the included generated rows retrieve less exact labeled evidence. The appendix reports Hit@1/3/5/10 and MRR@10 from full top-10 retrieval runs.

The extra evidence is useful but incomplete. Snap-HyRE converts a +9.5pp Hit@5 and +4.7pp MRR@5 retrieval shift into +3.9pp average BarExamQA answer accuracy over raw-question RAG, while exact gold evidence still appears in only about one in nine top-five lists. The remaining gap points to ranking, option mapping, and evidence use.

Table 3. Main answer accuracy (%). Bold marks the best non-oracle row within each dataset/model slice; averages are descriptive over available cells.

Method	BarExam 8B	26B	70B	Avg.	Housing 8B	26B	70B	Avg.
LLM	57.3	80.8	78.7	72.3	55.4	—	44.8	50.1
Raw question RAG	54.5	78.0	74.6	69.0	62.3	—	62.1	62.2
HyDE	56.1	80.3	80.2	72.2	59.1	—	62.2	60.7
Snap-HyRE (ours)	56.9	82.0	79.7	72.9	59.0	—	59.6	59.3
Gold	60.0	78.6	79.2	72.6	64.3	—	67.3	65.8

 Table 4. Evidence exposure at $k = 5$ (%). Generated-query rows average available model-specific generators; HousingQA uses state-filtered retrieval.

Method	BarExam Hit@5	BarExam MRR@5	Housing Hit@5	Housing MRR@5
Raw question RAG	1.4	0.7	37.0	23.3
HyDE	10.0	5.0	30.9	19.7
Snap-HyRE (ours)	10.9	5.4	26.3	16.6

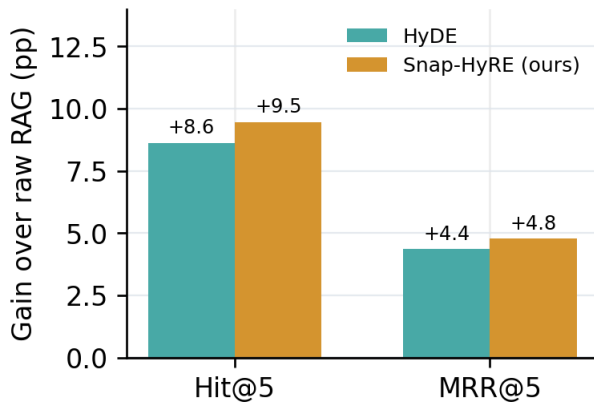


Figure 2. Mean BarExamQA retrieval gains over raw-question RAG. Generated search passages improve both top-five exposure and rank.

5.3. How the Win Happens

The case study shows the intended design pattern. The raw query is dominated by surface accident facts. HyDE moves toward legal language but remains broad. Snap-HyRE first forms the private draft judgment that the later fall is foreseeable because the original injury required crutches; retrieval from that frame surfaces compact proximate-cause rule passages. The final answerer still sees only the original question and retrieved legal text, so the win is a retrieval-query win, not answer leakage. Appendix A.4 lists the compact worked examples used to sanity-check this mechanism.

5.4. Cost and Context

Snap-HyRE and HyDE both add a first call before retrieval, so their end-to-end footprint is two logical model calls. The answer-stage token logs ask a nar-

rower question: after retrieval, how much text does the scored answer call process? On rows with token logs, Snap-HyRE averages 1,762 answer-stage tokens per question, compared with 2,634 for raw-question RAG and 2,442 for HyDE. The query-generation call is separate and reported in Appendix 10.

5.5. Evidence Is Not the Only Bottleneck

Oracle rows show what remains after retrieval is removed. On BarExamQA, gold passage accuracy is only modestly above the best generated-query rows, so option mapping and evidence use still bind. On HousingQA, gold passages remain far above LLM-only, showing headroom when the correct statute is supplied. Appendix A.2 reports the oracle and rewrite controls.

6. What Snap-HyRE Does and Does Not Fix

Snap-HyRE is a fixed generated-query intervention, not a universal replacement for raw retrieval. It is strongest on BarExamQA, where draft-guided legal passages improve both Hit@5 and final answer accuracy. HousingQA marks the boundary: once retrieval is scoped to the correct state, raw questions can already be strong search queries.

The failures are stage-specific: a draft can choose the wrong frame, a passage can drop anchors, and an answer model can misuse good evidence. Legal RAG papers should report retrieval exposure, answer use, and corpus-interface assumptions. Snap-HyRE helps when a question must be rewritten into authority language before retrieval begins.

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A. Appendix

A.1. Result Inclusion Policy

A row is used as a paper claim only when the result record names the dataset, provider, method, answer depth, metric values, and any row note needed to interpret the result. Rows without validated full results are omitted from the main comparison rather than imputed.

- Main benchmarks: BarExamQA and HousingQA.
- Providers: groq-llama8b, or-gemma4-26b, groq-llama70b.
- Main modes: llm_only, rag_simple, rag_hyde, snap_hyre, and golden_passage. rag_rewrite is a supplemental generated-query control, and golden_plus_neighbors is retained only as a neighbor-dilution diagnostic.
- Answer depth: $k = 5$ for retrieval rows.
- HousingQA interface: retrieval rows must use the jurisdiction state filter to count as main-interface rows.

Table 5. Coverage in the BarExamQA/HousingQA matrix. Included rows support paper claims; archived rows are older or unfiltered runs kept outside the main comparison.

Dataset	Included	Archived	Pending	Expected
BarExamQA	21	0	0	21
HousingQA	10	5	0	21
Total	31	5	0	42

A.2. Supplemental Controls

These tables expand the main claims with dataset-level anatomy, structured rewrite controls, oracle controls, and scored-answer usage accounting. The usage table reports answer-call accounting only; first-stage query-generation usage is logged separately.

A.3. Top-k Retrieval Diagnostics

BarExamQA generated passages raise the chance of seeing the labeled evidence throughout the top-10 list, although MRR remains modest because the exact supporting passage is rarely ranked first. HousingQA behaves differently: the jurisdiction filter is the major retrieval correction, and raw state-filtered questions already retrieve relevant evidence often.

A.4. Worked Examples

Table 6. Dataset-level result anatomy. Deltas are mean percentage-point differences over included row pairs where both methods exist.

Dataset	Raw evidence regime	Snap vs. raw pp	HyDE vs. Snap pp	Gold vs. raw pp	Interpretation
BarExamQA	Very low raw evidence exposure; generated legal text supplies missing vocabulary.	+3.8 (n=3)	-0.7 (n=3)	+3.6 (n=3)	Snap-HyRE clearest here.
HousingQA	Jurisdiction must be fixed before retrieval methods are compared; available state-filtered rows favor raw questions.	-2.9 (n=2)	+1.4 (n=2)	+3.6 (n=2)	Jurisdiction scope changes the comparison.

Table 7. Snap-HyRE compared with non-oracle controls. Cells are accuracy percentages; bold marks the strongest non-oracle row for that dataset/model slice. Rewrite is supplemental.

Dataset	Model	LLM	Raw question RAG	HyDE	Snap-HyRE	Rewrite	Snap vs. raw pp	Best main non-oracle
BarExamQA	Llama 3.1 8B	57.3	54.5	56.1	56.9	57.3	+2.4	LLM (57.3)
BarExamQA	Gemma 4 26B	80.8	78.0	80.3	82.0	80.7	+4.0	Snap-HyRE (ours) (82.0)
BarExamQA	Llama 3.3 70B	78.7	74.6	80.2	79.7	77.2	+5.1	HyDE (80.2)
HousingQA	Llama 3.1 8B	55.4	62.3	59.1	59.0	—	-3.3	Raw question RAG (62.3)
HousingQA	Llama 3.3 70B	44.8	62.1	62.2	59.6	—	-2.5	HyDE (62.2)

Table 8. Gold-passage oracle controls for the main datasets. Accuracy values are percentages; deltas are percentage points over included row pairs. Neighbor-augmented oracle rows are left to the appendix figure because they diagnose context dilution rather than the main retrieval claim.

Dataset	Model	Raw question RAG	Snap-HyRE	Gold	Gold vs. raw pp
BarExamQA	Llama 3.1 8B	54.5	56.9	60.0	+5.5
BarExamQA	Gemma 4 26B	78.0	82.0	78.6	+0.6
BarExamQA	Llama 3.3 70B	74.6	79.7	79.2	+4.6
HousingQA	Llama 3.1 8B	62.3	59.0	64.3	+2.0
HousingQA	Llama 3.3 70B	62.1	59.6	67.3	+5.2

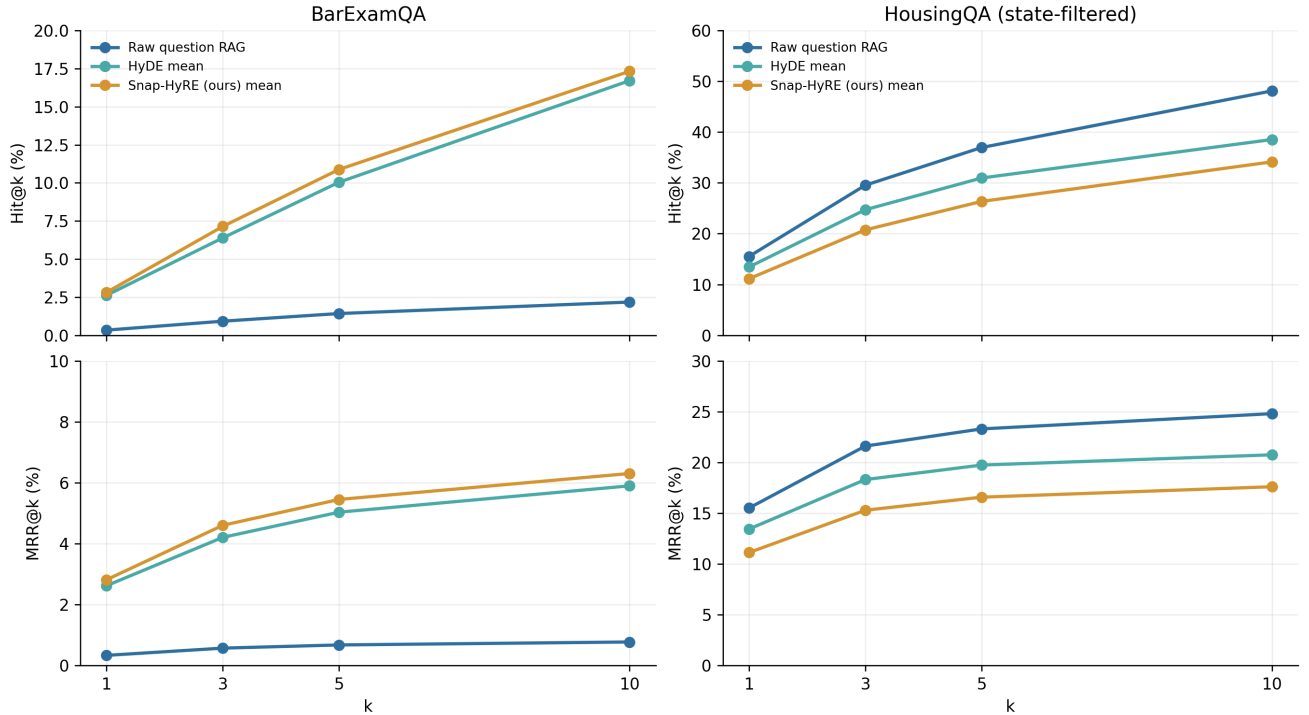


Figure 3. Top-k retrieval exposure and rank quality for full top-10 caches. BarExamQA generated-query curves average over the three model-specific query generators; HousingQA uses state-filtered retrieval caches.

Table 9. Supplemental neighbor-dilution diagnostic. Values are answer accuracy percentages for oracle rows only; positive deltas mean adding nearby retrieved legal text helped relative to the gold passage alone.

Dataset	Model	Gold	Gold + neighbors	Δ pp
BarExamQA	Llama 3.1 8B	60.0	61.8	+1.8
BarExamQA	Gemma 4 26B	78.6	80.7	+2.1
BarExamQA	Llama 3.3 70B	79.2	77.8	-1.4

Table 10. Token and scored-answer call accounting for included BarExamQA/HousingQA answer rows. Values cover rows with token fields and count only the call that produces the submitted answer from the original question and any supplied context, including format retries. First-stage query-generation usage is not included, so conceptual end-to-end call counts remain those in Table 1.

Method	Cells	Tok./q	Input tok./q	Output tok./q	Scored calls/q	Correct / 1M tok.
LLM	5	368	131	237	1.00	1487.2
Raw question RAG	5	2634	2295	339	1.00	241.5
HyDE	5	2577	2237	340	1.00	244.5
Snap-HyRE (ours)	4	2377	2001	376	1.00	268.3
Gold	5	1661	1447	215	1.00	404.5
Rewrite	3	2524	1969	555	2.00	284.2

Table 11. HousingQA raw retrieval with and without the jurisdiction metadata filter. Values are percentages from complete top-10 caches.

Corpus scope	Hit@1	Hit@3	Hit@5	Hit@10	MRR@10
National corpus	0.8	1.9	2.8	5.1	1.8
Jurisdiction metadata filter	15.5	29.5	36.9	48.1	24.8

Table 12. Top- k retrieval diagnostics from complete top-10 caches. Values are percentages. BarExamQA includes raw-question retrieval plus each model-specific generated-query cache; mean rows average the three generated-query models. HousingQA rows use the state-filtered retrieval interface and include only complete full-corpus caches.

Dataset	Model/scope	Method	n	Hit@1	Hit@3	Hit@5	Hit@10	MRR@10
BarExamQA	Shared	Raw question RAG	1195	0.3	0.9	1.4	2.2	0.8
BarExamQA	Mean over 3 models	HyDE	1195	2.6	6.4	10.0	16.7	5.9
BarExamQA	Mean over 3 models	Snap-HyRE (ours)	1195	2.8	7.1	10.9	17.3	6.3
BarExamQA	Llama 3.1 8B	HyDE	1195	2.8	5.5	8.3	13.5	5.2
BarExamQA	Llama 3.1 8B	Snap-HyRE (ours)	1195	2.3	6.3	9.5	14.8	5.4
BarExamQA	Gemma 4 26B	HyDE	1195	2.5	7.1	11.4	19.1	6.4
BarExamQA	Gemma 4 26B	Snap-HyRE (ours)	1195	3.2	7.9	12.1	18.7	6.9
BarExamQA	Llama 3.3 70B	HyDE	1195	2.6	6.5	10.5	17.6	6.1
BarExamQA	Llama 3.3 70B	Snap-HyRE (ours)	1195	3.0	7.2	11.0	18.5	6.6
HousingQA	State-filtered shared	Raw question RAG	6853	15.5	29.5	36.9	48.1	24.8
HousingQA	Mean over 2 full models	HyDE	6853	13.4	24.7	30.9	38.5	20.7
HousingQA	Mean over 2 full models	Snap-HyRE (ours)	6853	11.1	20.7	26.3	34.1	17.6
HousingQA	State-filtered Llama 3.1 8B	HyDE	6853	11.2	21.6	27.0	34.5	17.9
HousingQA	State-filtered Llama 3.1 8B	Snap-HyRE (ours)	6853	12.9	23.4	29.6	37.0	19.9
HousingQA	State-filtered Llama 3.3 70B	HyDE	6853	15.7	27.9	34.9	42.6	23.6
HousingQA	State-filtered Llama 3.3 70B	Snap-HyRE (ours)	6853	9.3	18.1	23.1	31.3	15.3

Table 13. Worked BarExamQA example (mbe_1175, Gemma 4 26B). The question describes a negligent driver, a broken leg, crutches, a banana peel, and a later fall. The gold answer is that the driver remains liable for both injuries.

Method	What changes
Raw question RAG	Search stays at accident surface facts: driver, crutches, banana peel, market fall. Retrieval returns generic accident and intervening-cause material; the gold passage is not retrieved. Predicts D; wrong.
HyDE	Generated passage moves toward proximate cause but stays broad. Retrieval still misses the gold rule passage. Predicts D; wrong.
Snap-HyRE (ours)	Private draft identifies the hinge: the first injury made the crutch-related fall foreseeable. The search passage retrieves MBE rule passages on proximate cause and foreseeable subsequent injury. Predicts B; correct.

A.5. Probe-only Exemplar Details

The retrieval-passage exemplar probe adds one fixed, sanitized dataset-specific real-passage-style exemplar to the generation prompt without providing row-specific answer evidence, answer labels, or document ids. On the 20-question Gemma 4 26B retrieval slices, the exemplar improves over base Snap-HyRE on the two main datasets: BarExamQA moves from Hit@5/MRR@5 0.05/0.03 to 0.15/0.09, and HousingQA moves from 0.05/0.05 to 0.10/0.10. These are query-shape diagnostics rather than fixed-method results.

A.6. Reproducibility Statement

All tables and figures are derived from the result records used for this draft. The released package will include the records, plotting code, and paper sources needed to reproduce the reported matrices and figures.

Table 14. Probe-only 20-question retrieval comparison for Gemma 4 26B with one fixed real-passage exemplar per dataset. Cells are Hit@5/MRR@5 percentages. These rows are not part of the main matrix.

Dataset	Raw	Snap-HyRE	Exemplar
BarExamQA	0.0/0.0	5.0/2.5	15.0/8.5
HousingQA	0.0/0.0	5.0/5.0	10.0/10.0